CS23331-DAA-2025-CSE / 1-DP-Playing with Numbers

1-DP-Playing with Numbers

Started on	Wednesday, 30 September 2026, 8:31 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 8:55 AM
Time taken	23 mins 32 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00 Flag question

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*

Output:*6*

Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

unsigned long long countways(int n){
    if(n<=0) return 0;
    if(n==1 || n==2) return 1;
    if(n==3) return 2;

    unsigned long long dp[n+1];
    dp[1] = 1;
    dp[2] = 1;
    dp[3] = 2;

    for(int i=4; i<=n; i++){
        dp[i] = dp[i-1] + dp[i-3];
    }
    return dp[n];
}

int main(){
    int n;
    scanf("%d", &n);
    unsigned long long ans = countways(n);
    printf("%llu", ans);
    return 0;
}
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

2-DP-Playing with chessboard

Started on	Wednesday, 23 September 2026, 9:11 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 8:22 AM
Time taken	6 days 23 hours
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00 Flag question

Playing with Chessboard:

Ram is given with an n*n chessboard with each cell with a monetary value. Ram stands at the (0,0), that the position of the top left white rook. He is been given a task to reach the bottom right black rook position (n-1, n-1) constrained that he needs to reach the position by traveling the maximum monetary path under the condition that he can only travel one step right or one step down the board. Help ram to achieve it by providing an efficient DP algorithm.

Example:

Input

3
1 2 4
2 3 4
8 7 1

Output:

19

Explanation:

Totally there will be 6 paths among that the optimal is
Optimal path value:1+2+8+7+1=19

Input Format

First Line contains the integer n
The next n lines contain the n*n chessboard values

Output Format

Print Maximum monetary value of the path

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>

int max(int a, int b){
    return (a >= b) ? a : b;
}

int main(){
    int n;
    scanf("%d", &n);
    if(n==0)
        return 0;

    int arr[n][n];
    for(int i=0; i<n; i++){
        for(int j=0; j<n; j++){
            scanf("%d", &arr[i][j]);
        }
    }

    int dp[n][n];

    dp[0][0] = arr[0][0];
    for(int i=1; i<n; i++)
        dp[0][i] = arr[0][i] + dp[0][i-1];
    for(int i=1; i<n; i++)
        dp[i][0] = arr[i][0] + dp[i-1][0];

    for(int i=1; i<n; i++){
        for(int j=1; j<n; j++)
            dp[i][j] = arr[i][j] + max(dp[i-1][j], dp[i][j-1]);
    }

    int ans = dp[n-1][n-1];
    printf("%d", ans);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 2 4 2 3 4 8 7 1	19	19	✔
✔	3 1 3 1 1 5 1 4 2 1	12	12	✔
✔	4 1 1 3 4 1 5 7 8 2 3 4 6 1 6 9 0	28	28	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

3-DP-Longest Common Subsequence

Started on	Wednesday, 23 September 2026, 8:27 AM
State	Finished
Completed on	Wednesday, 23 September 2026, 9:08 AM
Time taken	41 mins 1 sec
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1	a	g	**g**	**t**	**a**	**b**	
s2	**g**	x	**t**	x	**a**	y	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```c
#include<stdio.h>
#include<string.h>

int longestsubsequence(char s1[], char s2[]){
    int n1 = strlen(s1);
    int n2 = strlen(s2);

    int dp[n1+1][n2+1];

    for(int i=0; i<=n1; i++){
        for(int j=0; j<=n2; j++){
            if(i == 0 || j == 0)
                dp[i][j] = 0;

            else if(s1[i-1] == s2[j-1])
                dp[i][j] = dp[i-1][j-1] + 1;

            else
                dp[i][j] = (dp[i][j-1] >= dp[i-1][j]) ? dp[i][j-1] : dp[i-1][j];
        }
    }
    return dp[n1][n2];
}

int main(){
    char s1[100];
    char s2[100];
    scanf("%s", s1);
    scanf("%s", s2);

    int ans = longestsubsequence(s1, s2);
    printf("%d", ans);
    return 0;
}
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

CS23331-DAA-2025-CSE / 4-DP-Longest non-decreasing Subsequence

4-DP-Longest non-decreasing Subsequence

Started on	Wednesday, 30 September 2026, 8:56 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 9:20 AM
Time taken	24 mins 13 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00 Flag question

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int longest_non_decreasing_subsequence(int arr[], int n) {
    if (n == 0) return 0;

    int dp[n];
    int max_len = 1;


    for (int i = 0; i < n; i++) {
        dp[i] = 1;
    }
    for (int i = 1; i < n; i++) {
        for (int j = 0; j < i; j++){
            if (arr[j] <= arr[i] && dp[j] + 1 > dp[i]) {
                dp[i] = dp[j] + 1;
            }
        }
        if (dp[i] > max_len) {
            max_len = dp[i];
        }
    }

    return max_len;
}

int main() {
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    printf("%d\n", longest_non_decreasing_subsequence(a, n));

    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.